User Interface & User Experience



PeTIK Jombang

Pesantren PeTIK II YBM PLN
Jl. KH. Bisri Syansuri RT/01 RW/05, Plosogeneng,
Kec. Jombang, Kabupaten Jombang, Jawa Timur

YBM PLN
Yayasan Baitul Maal

PeTIK Jombang

YBM PLN
Yayasan Baitul Maal

5. Desain UI Berbasis Web dan Mobile

User Interface

- Process designers to build interfaces, focusing on looks or style.
- Designers create interfaces which users find easy to use and pleasurable.
- UI design refers to graphical user interfaces and other forms (voice)

UI Website vs UI Mobile

- Desain UI Website lebih kompleks
- Memperhatikan layar setiap device: laptop, tablet, dan smartphone.
- Layout website:
- Layout large (>992px)
- Layout medium (>768px)
- Layout small (576px)
- Demo UI Website

- Desain UI Mobile tidak sekompleks desain website
- Hanya menyesuaikan satu device: smartphone
- Demo UI Mobile

Mobile

- Aplikasi mobile merupakan aplikasi yang dibangun untuk platform tertentu - IOS atau Android.
- Walaupun saat ini sudah terdapat teknologi yang mampu mengembangkan aplikasi yang berjalan di multi platform sekaligus.
- Aplikasi mobile perlu didownload dan diinstall melalui playstore
- Memiliki kebebasan dalam mengakses sumber daya sistem
- Aplikasi mobile hanya berjalan di perangkat sendiri

Mobile

Kelebihan:

- Lebih cepat dibandingkan web
- Dapat bekerja secara offline
- Fungsionalitas lebih besar dalam mengakses sumber daya sistem

Kekurangan:

- Lebih mahal untuk dibuat daripada web
- Kompatibilitas terbatas hanya untuk masing-masing platform
- Sulit dan Mahal dalam melakukan maintain dan update

Website

- Aplikasi website adalah aplikasi yang diakses di internet melalui browser.
- Hanya membutuhkan aplikasi browser
- Dapat diakses dari platform apapun, tidak seperti mobile.
- Aplikasi website tidak perlu didownload dan diinstall seperti aplikasi mobile
- Aplikasi website dapat menerapkan Responsive Design sehingga tampilannya dapat terlihat seperti aplikasi mobile.

Website

Kelebihan:

- Tidak perlu diinstall - berjalan di browser
- Mudah dimantain
- Tidak perlu di-update
- Lebih cepat dibuat dibandingkan mobile

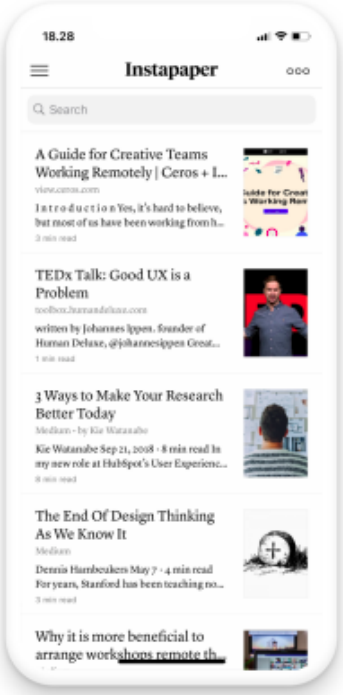
Kekurangan:

- Terbatas untuk bekerja secara offline
- Lebih lambat dari mobile
- Fungsionalitas lebih kecil dalam mengakses sumber daya sistem

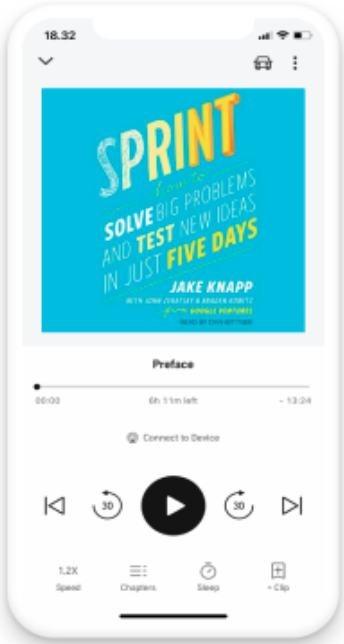
Basic Principles

- Strive for minimalism
- Place your most important elements near the bottom or center
- Reduce the need for typing
- Increase the size of your tap targets
- Prioritize legibility
- Give immediate and good feedback
- Enable easy error recovery

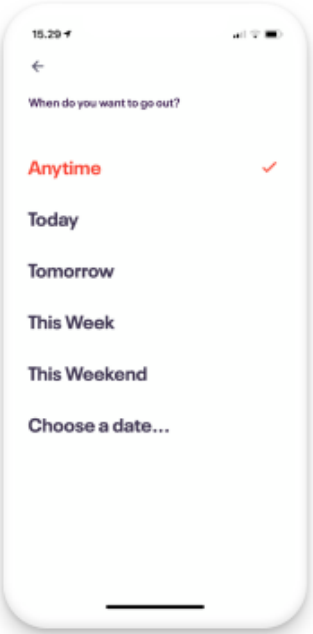
Basic Principles



Strive for minimalism

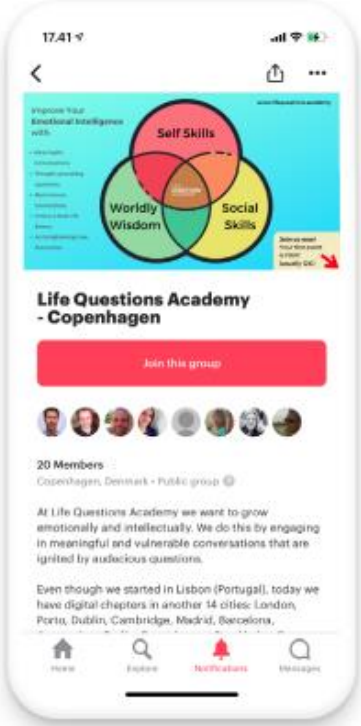


Place important elements near the bottom or center

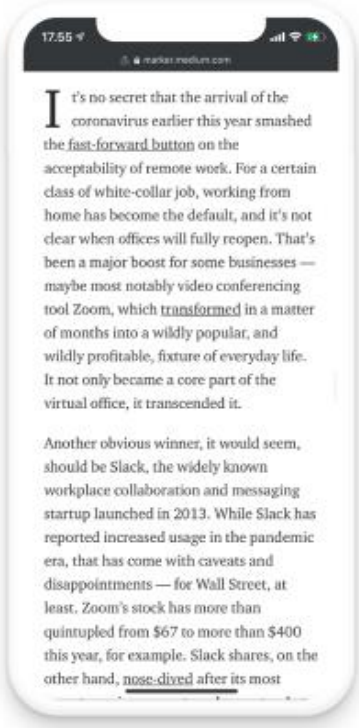


Reduce the need for typing

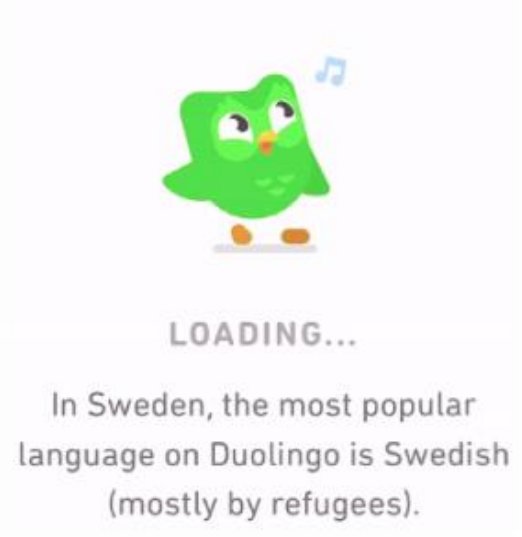
Basic Principles



Increase the size of tap targets



Prioritize legibility



Give immediate and good feedback

Mobilize, don’t Miniaturize

- Mobilizing means reconsidering the entire purpose of the application, not just changing display technologies or interaction nuances.
- This concept, to rethink what is desirable and possible for the mobile environment and to build and rebuild accordingly
- Concept:
- The Carry Principle: Device must be small enough so that it can be readily carried
- Small Device
- Specialized Multi-Purpose
- Personal Device: like a wallet, if loss will be noticed and rectified quickly.
- Customized Device
- Always On, Always Connected
- Battery-Powered
- Inconsistent Connectivity

User Context

- Mobile device contexts are more varied, and more difficult to predict and discover.
- Mobile devices share with ubiquitous computing the ability to discern user context.
- The mobile device knows nothing about the environment but has the resources and features that could enable it to learn much about its user.
- Myriad sources of information are possible: GPS, Wi-Fi, Bluetooth, Sensors ,Calendars, Cameras, Local Data Source
- When these information are combined, they can give the users enormous benefits

Handling Device Proliferation

Handling device proliferation is a necessity.

Four approaches to designing an application run on multiple devices:

- Targeted: select a set of targeted devices then write an application that works
- Least common denominator: Select technologies and designs that will work on all device
- Automatic translation: use a technology that converts some standard core function
- Class-based: identify groups of devices with common use and rendering characteristics

Emulators and Simulators

Emulators are a cheap and easy way to test applications. Simulators are slightly more suspect.

Developers and designers tend to rely on simulators and emulators to test their applications

Purpose:

.Emulators and simulators can be used by developers for interactive debugging of logic

.Can be used for usability testing to understand components of architecture

.Do not use for system testing, unit testing, or user acceptance testing.

Detail Design Recommendations

Design recommendations have to focus on a particular platform and set of devices.

Good resource for design recommendations (style guide):

- Platform Providers
- Standards Organizations
- Carriers and Device Manufacturers
- Third-Party Guidelines

Basic Principles

- Easy Navigation
- Responsive design: Design of a website that fits well with all the smartphones, computers, laptops or any display devices irrespective of their aspect ratios.
- Same Color Scheme (Consistency)
- Comfortable UI: If the UI is good then the user would definitely like to spend more time on it. It is the job of the designer to make the UI look fresh and clear
- Contents meet user goals for visiting websites
- Performance: Quick to show something

Basic Principles

.Feedback about progress: Feedback is a response about the task, process or event after its completion or during its stage of completion. Know what's going on inside the system or website in this case

.Avoid "Alert/Dialogs" when not necessary: Don't annoy the user's by providing unnecessary dialogs every time when they do something on the website.

.Mitigate 404 or 500 errors: try to design a good 404 error page

UI Design Pattern

- User interface (UI) design patterns are good solutions to standard user interface design problems.
- Patterns have become a good method for a new user interface designer to learn good, well-practiced solutions
- UI patterns provide a good starting point for specific parts of an application.

About User Interface Pattern

- A design pattern documents known good solutions to frequently occurring design problems.
- User interface design patterns are generally identified and articulated by design experts.
- There are three types of UI design pattern: patterns of practice, user interface design structures, and corporate patterns.
- Patterns of practice are closer to best practices in development
- User interface design patterns are solutions that likely work across a wide range of applications and on different platforms

About User Interface Pattern

- Mobilization:
- Patterns targeted at the mobile space must take into account the varying capabilities and user interface styles of the native operating system
- Universal Patterns
- Pure version of user interface design patterns, and apply to a wide variety of applications and across platforms.
- Corporate Patterns (Library)
- Many organizations standardize their design process using style guides, and pattern library.

Screen Design

The screen design patterns apply to the design of individual screens in myriad situations.

Screen Design Pattern:

- List-based Layout: Mobile devices vary in their screen dimension ratios as well as size. Some have a longer horizontal dimension; others are vertical or close to square
- Table-based Layout: Many devices have a launch screen, with two or three columns of icons, from which major components can be started.
- Location Selection: This is a generic interface to obtain, save, and manage the user's location across a variety of applications and device capabilities.

Screen Design

Screen Design Pattern :

- Returned Results
- Menu: A menu is a list of commands. It can be the main screen of an application, or a set of commands applicable to an item or part of the application.
- Tab Navigation: Tabs are a common mechanism used to arrange more controls than can fit on a single page.
- Breadcrumbs: Breadcrumbs are a popular mechanism for locating the user within a site and providing supplemental navigation.

Application Navigation

Application navigation is the first place where developers from the desktop world get in trouble

These patterns address some of the common navigation issues faced by mobile applications:

- List Navigation: Navigating between items in a list quickly causes navigation challenges.
- Game Navigation: While games vary greatly, the navigation structure to support the game should not.
- Alphabetic Listings – Short: A list of alphabetically sorted items can be navigated more quickly using a combination of numbered links and fetching.

Application Navigation

- Alphabetic Listings – Long: A list of alphabetically sorted items can be navigated more quickly using a combination of numbered links and fetching.
- Softkey and Button Management

PeTIK Jombang

YBM PLN
Yayasan Baitul Maal

TERIMA KASIH
ATAS SEGALA PERHATIAN
SEMOGA BERMANFAAT...

UI/UX